



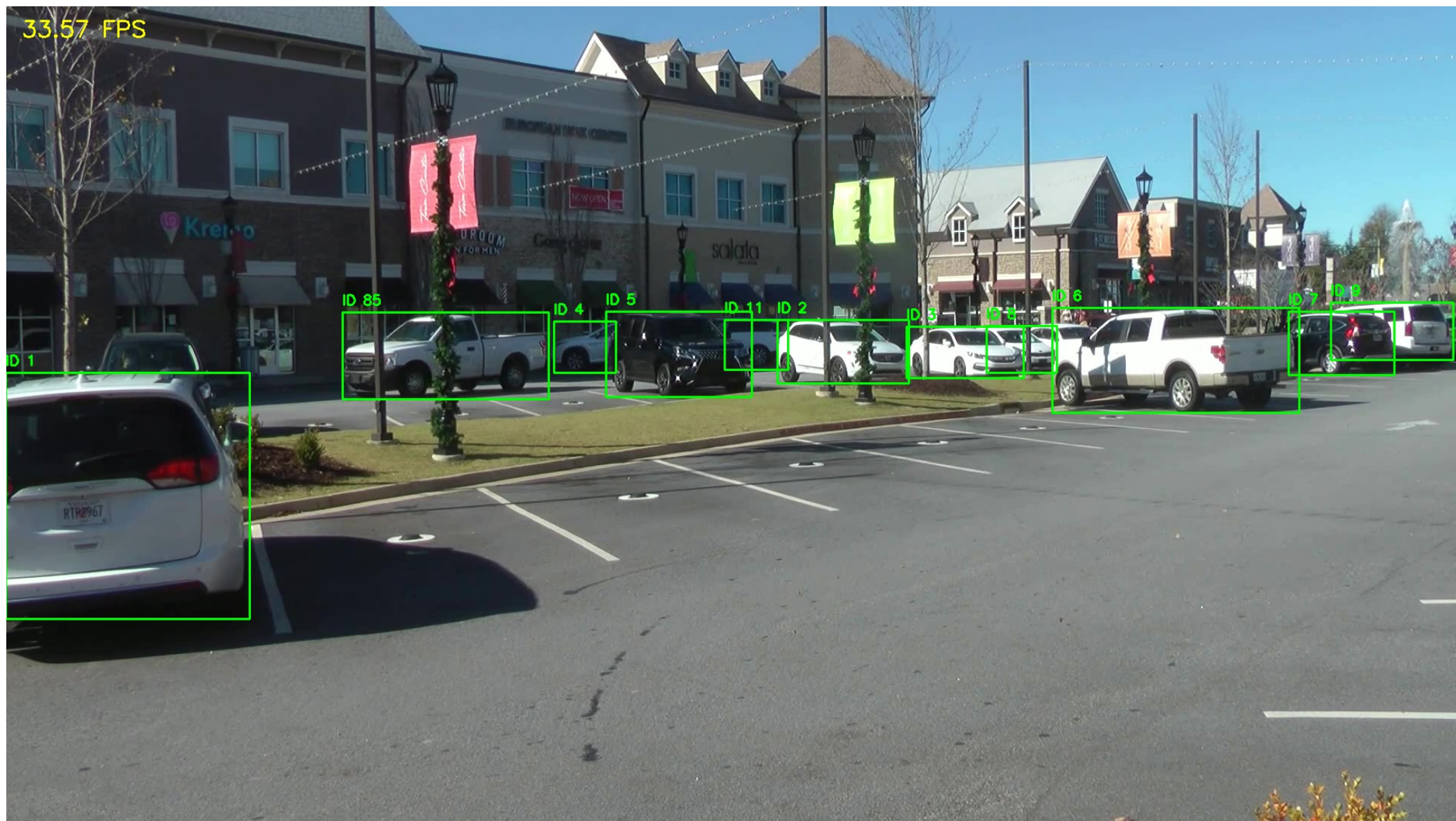
A Lightweight Spatio-Temporal Framework for Repairing Fragmented Vehicle Tracks

Brian Guo¹ Dr. Xinyu Liu²
University of Michigan - Ann Arbor

Introduction / Background

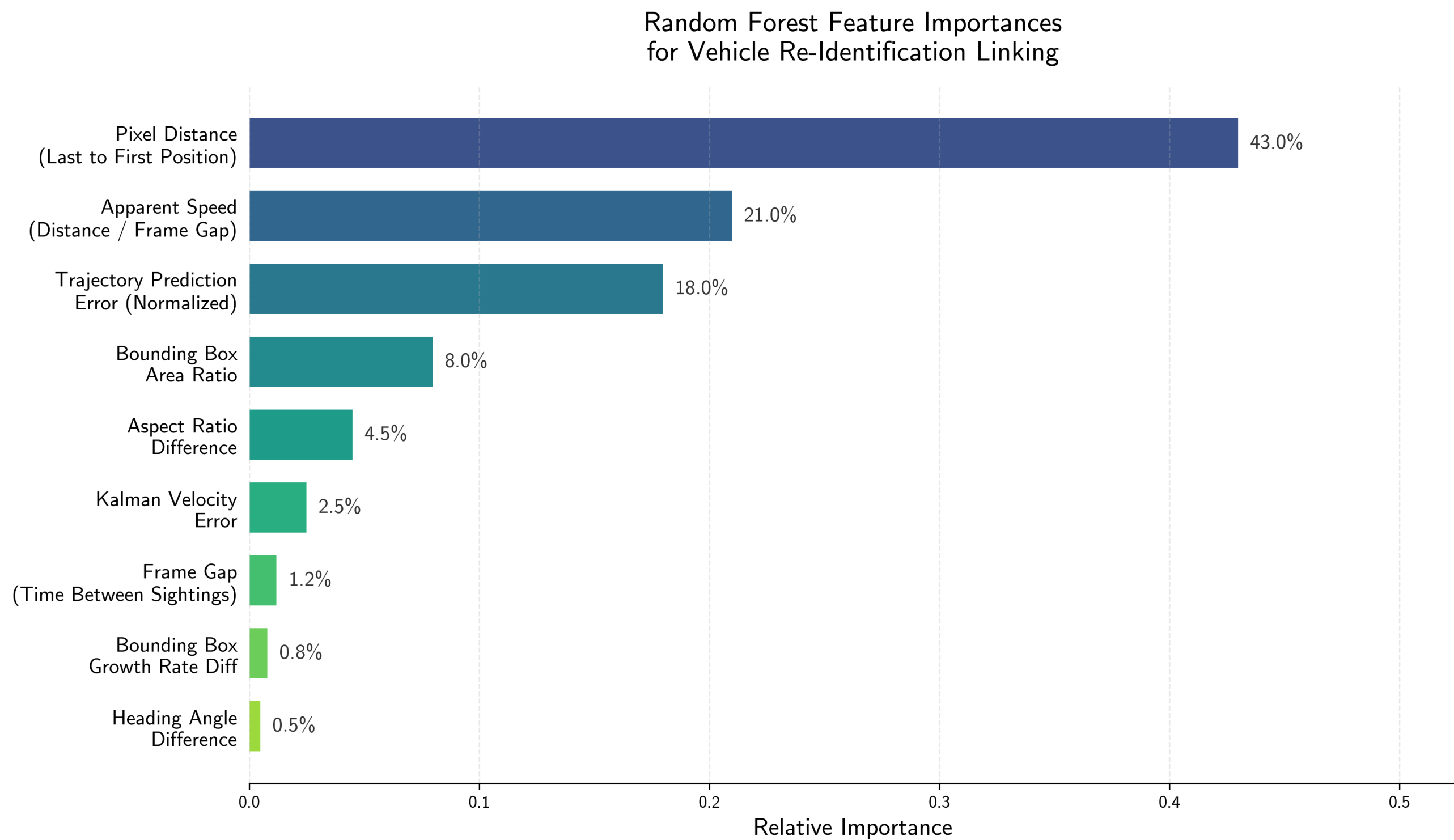
Computer vision (CV) methods like YOLO and DeepSORT have enabled automatic vehicle detection and tracking using video sequences from complex traffic environments, which is often classified as a Multi-Object Tracking (MOT) task. Despite these innovations, the methods may fail in the face of prolonged occlusions or dense traffic interactions.

This research proposes a lightweight computational framework to recover the physical vehicle trajectories, by correcting missed or swapped tracklets produced by traditional CV models. We designed and implemented the entire pipeline, from track splitting and feature engineering to model training.



Methodology

- Data:** Parking lot and airport footage containing 140 unique DeepSORT track IDs across 31 labeled vehicles
- Split:** 936 positive and 2,956 negative pairs
- Feature Engineering:** The relinking algorithm operates on a random forest trained with 9 distinct features and 5-fold cross-validation.



Pipeline Overview

From raw YOLO and DeepSORT detections to analyst relink report.



Figure 1. Overview of the six-stage vehicle re-id pipeline from start to finish.

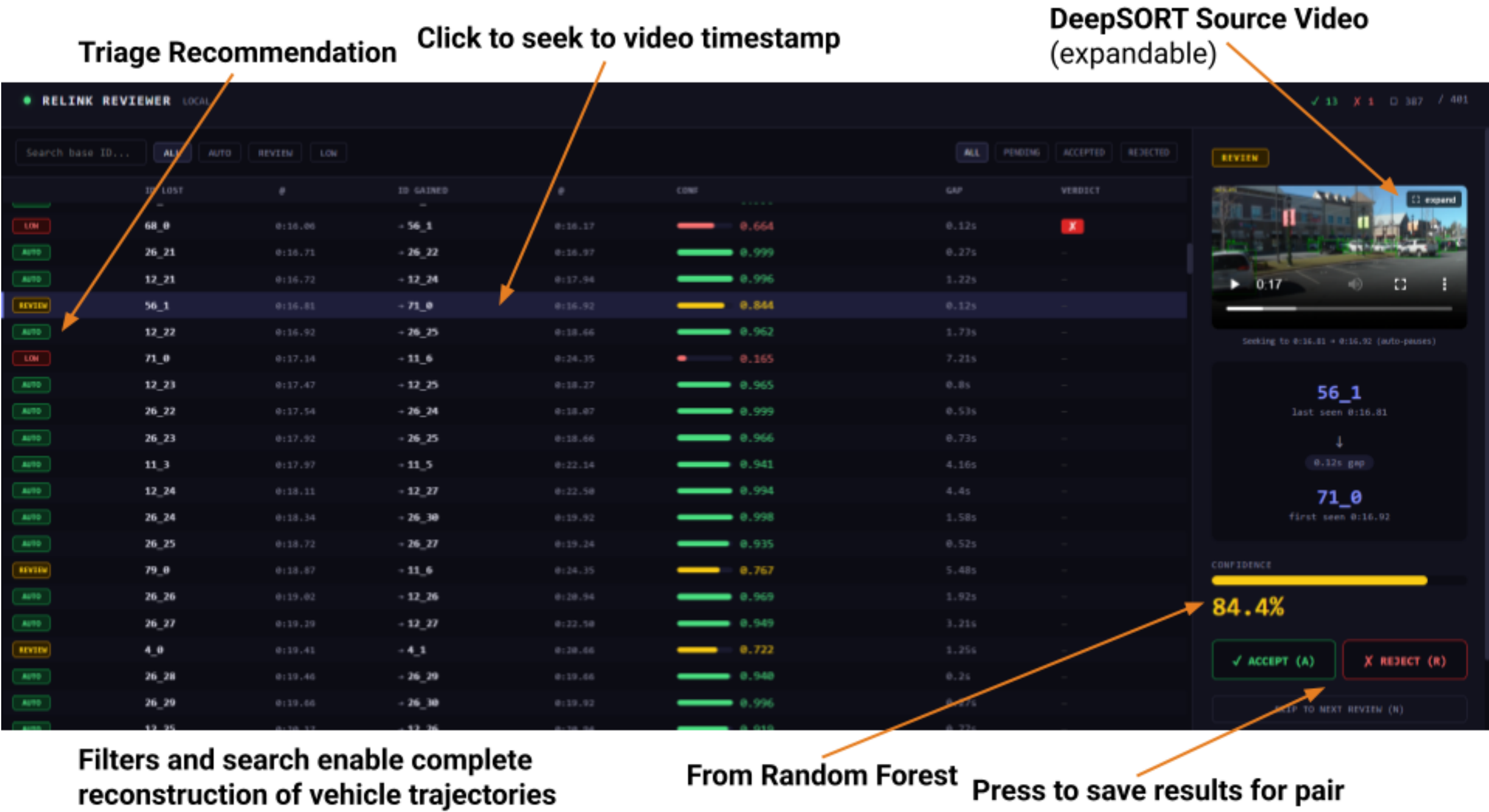
Findings & Preliminary Results

- The threshold (ex. $\theta = 0.40$) is the cutoff: any pair scoring above it gets classified as "same vehicle," anything below gets rejected.
- Precision-recall tradeoff remains stable across thresholds

Threshold	Precision	Recall	F1
0.30	0.81	0.90	0.85
0.35	0.82	0.89	0.86
0.40	0.83	0.88	0.86
0.50	0.84	0.87	0.85
0.60	0.86	0.82	0.84
0.70	0.89	0.79	0.84
0.80	0.92	0.72	0.81

Conclusion

Our pipeline takes in trajectory fragments produced by YOLO and DeepSORT as input, feeds them to the relinking algorithm, and proposes complete trajectories.



Real-World Impact: This research provides a promising way for practitioners such as airport operators and urban planners to extract valuable insights into traffic flow using research into CV pipelines.

Limitations & Future Work

Study Limitations:

- Ground-truth labels were created with human annotation, which can introduce subjective error into both training and evaluation.
- The model relies on spatial and temporal features, so vehicles that are partially visible make unreliable bounding boxes, making relinking difficult.

Next Steps:

- Human verification of proposed links are still needed since the model can have a high-confidence for an incorrect pair.
- Multi-modal LLMs with vision capabilities can compare snapshot pairs from the relink report and classify each proposed link as same vehicle, new vehicle, or no longer present.
- Early experiments with LLMs show promising accuracy, but they rely on the existing pipeline for candidate generation, cannot operate at frame-level scale, and adds significant inference cost.

References

